A Breif Introduction to Sphinx

Technical Documentation with Continuous Integration and Deployment

Shao-Ching Huang

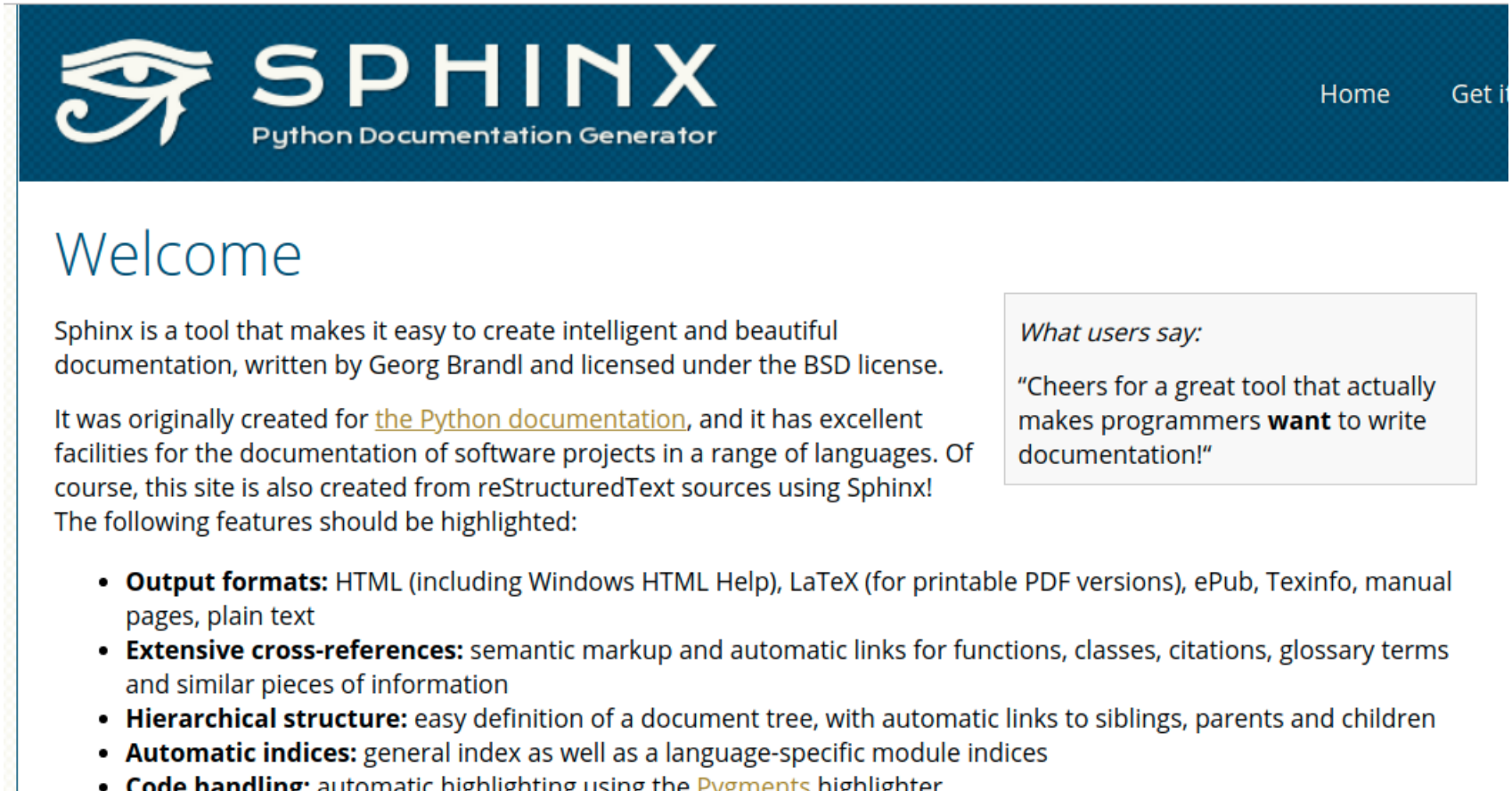
2019-10-31

Technical documentation system: considerations

- Not the same as writing a blog article
- Writing
  - Focus on contents and structures, strict and fixed formatting
  - Principle of separation: contents vs. layout
- Review
  - easy to see changes, revert, comment, collaborate etc
- Other considerations
  - Subject experts are not necessarily web developers
  - Multiple formats generated from the same source (e.g. web pages and pdf report/manual)

What is Sphinx

- Sphinx converts reStructuredText files into HTML websites and PDF, EPub, Texinfo and man.
- reStructuredText (RST, ReST, or reST) is a file format for textual data for technical documentation.
  - a more powerful markdown without diverging flavors



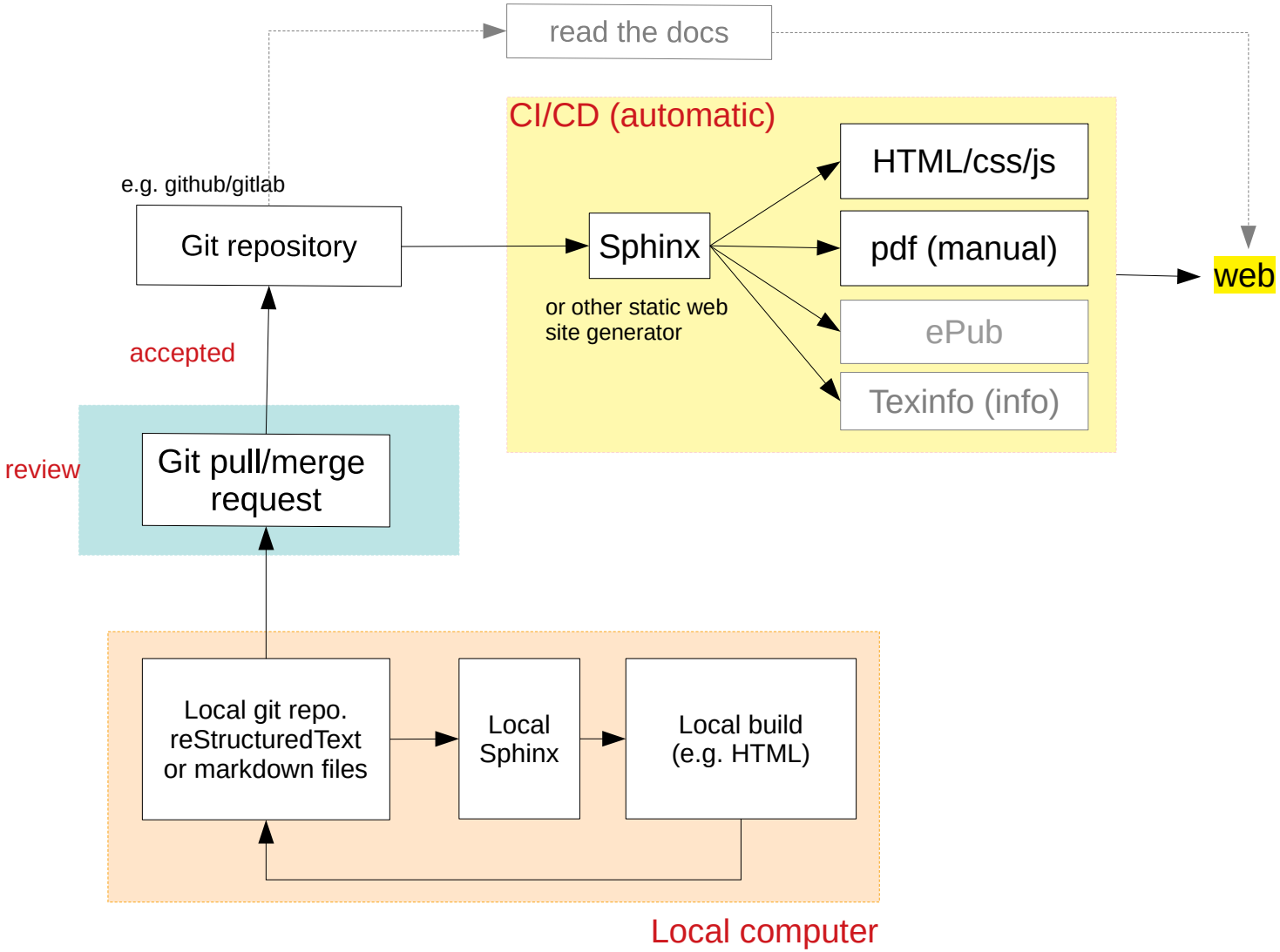
Advantages of using Sphinx

- Based on text files (close to plain text)
  - Use one's favoriate editor/local environment for writing
  - Easy to track changes using version control
  - Standard git pull (or merge) request review process
  - Easy to programmatically manipulate the text files, if needed
  - The entire workflow is open-source based, no vendor lock-in
- Generate static HTML pages
  - No security concerns
  - No need to maintain WordPress that often "automatically" breaks
  - Can use free hosting services (github, gitlab, read the docs, etc.) if desired

Sphinx demo

- view the source
  - `https://github.com/schuang/sphinx-tutorial`
- view the site
  - `https://schuang.github.io/sphinx-tutorial`

Using Sphinx with CI: the big picture



Continuous integration/deployment (CI/CD)

- A software engineering technique to integrate early and often to avoid "integration hell"
- Best practices:
  - Automate the build
  - Commit changes to repository regularly
  - Every commit is built and tested

Travis CI

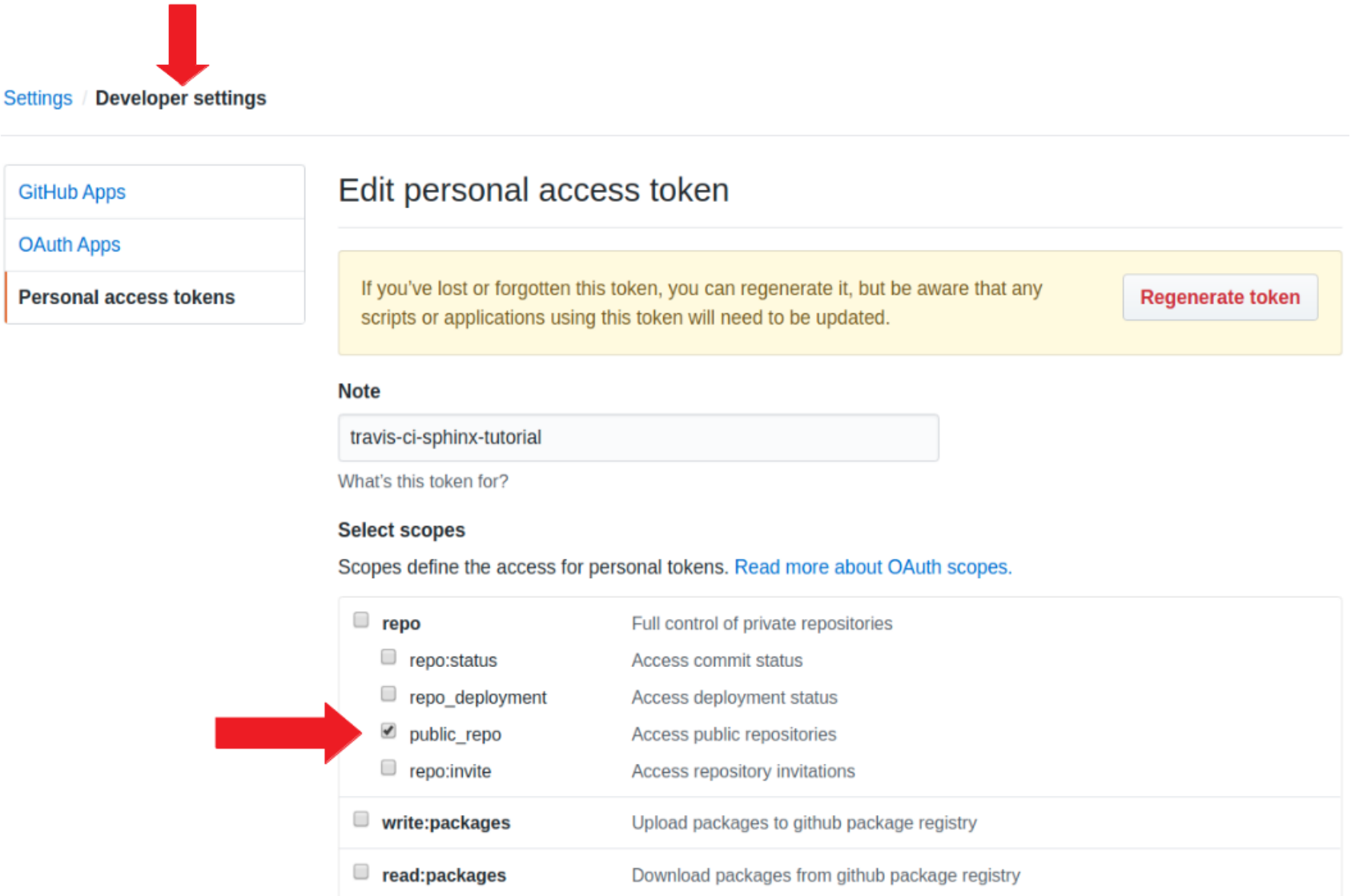
- Travis CI is a hosted contiguous integration service
- Integration with GitHub
- Free for open-source projects, paid plans for private repositories
- Web site: https://travis-ci.com/

There are also several other popular CI systems with a variety of build systems (C, C++, Python, PHP, etc.)

Enabling Travis CI on a Github repository

- Create an access token in github.com
- Past the access token into Travis CI to enable access to Github
  - e.g. `$GITHUB_TOKEN`
  - Use it in `.travis.yml`
- Other CI/CDs have a similar procedure, e.g. `gitlab.com`

github.com personal access token



travis-ci.com environment variable

Current Branches Build History Pull Requests Settings More options

General

Build pushed branches ?

Limit concurrent jobs ?

Build pushed pull requests ?

Auto Cancellation

Auto Cancellation allows you to only run builds for the latest commits in the queue. This setting can be applied to builds for Branch builds and Pull Request builds separately. Builds will only be canceled if they are waiting to run, allowing for any running jobs to finish.

Auto cancel branch builds

Auto cancel pull request builds

Environment Variables

Customize your build using environment variables. For secure tips on generating private keys read our documentation

GITHUB_TOKEN ••••••••••••••••• Available to all branches

If your secret variable has special characters like `&`, escape them by adding `\` in front of each special character. For example, `ma&w!doc` would be entered as `ma\&w\!doc`.

NAME | VALUE | BRANCH

Name | Value | All branches

DISPLAY VALUE IN BUILD LOG

Add

YAML to control the CI process I

```
$ cat .travis.yml
language: python
branches:
  only:
  - master
python:
- '3.6'
install:
- pip install -U pip
- pip install -r requirements.txt
script:
- make html
deploy:
  provider: pages
  skip_cleanup: true
```

YAML to control the CI process II

```yaml
github_token: $GITHUB_TOKEN
local_dir: build/html
on:
  branch: master
```

pip-install additional packages

```
$ cat requirements.txt

sphinx
sphinx_rtd_theme
recommonmark
```

Travis CI jobs

schuang / hoffman2-user-guide build passing

Current Branches Build History Pull Requests More options

master Update README.md #14 passed Restart build

Commit b75c300
Compare 8a86cee..b75c300
Branch master
Shao-Ching Huang

Ran for 8 min 15 sec
about 17 hours ago

Python: 3.6
AMD64

Job log View config

Remove log Raw log

```
1   Worker information                                                                  worker_info   0.07s
6                                                                                                     0.01s
7   Build system information                                                            system_info
159
160                                                                                                   5.82s
                                                                                        docker_mtu
                                                                                        resolvconf
161 $ git clone --depth=50 --branch=master https://github.com/schuang/hoffman2-user-guide.git schuang/hoffman2-user-guide   git.checkout   0.74s
165
166                                                                                                   0.01s
167 Setting environment variables from repository settings
168 $ export GITHUB_TOKEN=[secure]
169
170 $ source ~/virtualenv/python3.6/bin/activate                                                      0.01s
171 $ python --version
172 Python 3.6.7
173 $ pip --version
174 pip 19.0.3 from /home/travis/virtualenv/python3.6.7/lib/python3.6/site-packages/pip (python 3.6)
175 $ sudo apt-get -y install texlive-full                                              install.1     420.62s
```

Useful Sphinx settings in `conf.py`

To support Github pages:

```
extensions = [ 'sphinx.ext.mathjax',
               'sphinx.ext.githubpages' ]
```

To also support markdown source files:

```
source_suffix = { '.rst': 'restructuredtext',
                  '.md' : 'markdown' }
```

Travis CI demo

What to look:

- See the github repository and travis-ci.com
- Commit to github triggering Travis CI build job
- Watch the "job log" in real time
- Build status badge

gitlab pages pipeline

- similar to Travis CI
- use `.gitlab-ci.yml` (different syntax than `.travis.yml`)
- Can use arbitrary docker image from docker hub
- CI is ingegrated in gitlab, so no need to create the access token
- use `/public`

```
$ cat .gitlab-ci.yml
image: python:3.6

pages:
  script:
    - pip install -U pip
    - pip install -r requirements.txt
    - make html
    - mv build/html/ public/
  artifacts:
    paths:
      - public
  only:
    - master
```

gitlab pipeline jobs

GitLab Projects Groups More Search or jump to...

H Hoffman2 user guide

- Project
- Repository
- Issues 0
- Merge Requests 0
- CI / CD
  - Pipelines
  - Jobs
  - Schedules
  - Charts
- Security & Compliance
- Operations
- Packages
- Wiki
- Snippets
- Settings

Shao-Ching Huang › Hoffman2 user guide › Jobs › #333519076

passed **Job #333519076** triggered 1 day ago by **Shao-Ching Huang**

```
Running with gitlab-runner 12.3.0 (a8a019e0)
  on docker-auto-scale 0277ea0f
Using Docker executor with image python:3.5 ...
Pulling docker image python:3.5 ...
Using docker image sha256:8f3c1f9250c613d2a75bba6672ddfbc26e89f1a4147bdf50837074f278a24e33 for python:3.5 ...
Running on runner-0277ea0f-project-15003515-concurrent-0 via runner-0277ea0f-srm-1572033077-aab8660c...
Fetching changes with git depth set to 50...
Initialized empty Git repository in /builds/huangsc/hoffman2-user-guide/.git/
Created fresh repository.
From https://gitlab.com/huangsc/hoffman2-user-guide
 * [new ref]         refs/pipelines/91630618 -> refs/pipelines/91630618
 * [new branch]      master                  -> origin/master
Checking out 5a0f7675 as master...

Skipping Git submodules setup
$ pip install -U pip
Requirement already up-to-date: pip in /usr/local/lib/python3.5/site-packages (19.3.1)
$ pip install -r requirements.txt
Collecting sphinx
  Downloading https://files.pythonhosted.org/packages/8e/4c/95a21788db2e1653e931420f561015a0bbc9bd4660c4520467ab9e733eb2/Sphinx-2.2.0-
py3-none-any.whl (3.3MB)
Collecting sphinx_rtd_theme
  Downloading
https://files.pythonhosted.org/packages/60/b4/4df37087a1d36755e3a3bfd2a30263f358d2dea21938240fa02313d45f51/sphinx_rtd_theme-0.4.3-
py2.py3-none-any.whl (6.4MB)
Collecting sphinxcontrib-qthelp
  Downloading
https://files.pythonhosted.org/packages/ce/5b/4747c3ba98b3a3e21a66faa183d8f79b9ded70e74212a7988d236a6eb78a/sphinxcontrib_qthelp-1.0.2-
py2.py3-none-any.whl (90kB)
Collecting sphinxcontrib-serializinghtml
  Downloading
https://files.pythonhosted.org/packages/57/b3/3648e48fa5682e61e9839d62de4e23af1795ceb738d68d73bd974257a95c/sphinxcontrib_serializinghtm
l-1.1.3-py2.py3-none-any.whl (89kB)
Collecting sphinxcontrib-devhelp
  Downloading
https://files.pythonhosted.org/packages/b0/a3/fea98741f0b2f2902fbf6c35c8e91b22cd0dd13387291e81d457f9a93066/sphinxcontrib_devhelp-1.0.1-
py2.py3-none-any.whl (84kB)
Collecting babel!=2.0,>=1.3
```

pages

Retry

Duration: 1 minute 4 seconds

Timeout: 1h (from project)

Runner: shared-runners-manager-5.gitlab.com (#380986)

Job artifacts

Download Browse

Commit 5a0f7675

remove texlive since apt is not available

Pipeline #91630618 for master

test

→ pages

Summary I

- Sphinx is powerful documentation generator
  - Designed for technical documentation, used by many projects
  - Can easily include images, math, and code listing in style
  - Static web pages are light-weight, robust, and have no security issues
- Single source, multiple output formats
  - Separation of contents and display formatting/layout
  - Contents written in plain texts, formatting controlled by css/template/etc.
  - Ideal for git-based version control (pull/merge request etc.)
- Contiguous Integration (CI) automate rebuild when there is a change
  - Travis CI builds from Github sources, place HTML on Github pages

Summary II

- Total control of the documentation production process, including themes
- CI can be applied to other actions (by essentially arbrary scripts)

- Freedom of having different hosting options
  - Local web server (i.e. just copy the HTML directory) - total control
  - Github/Gitlab pages using contiguous integration (CI) - total control
  - Hosting service like Read the Docs (but have to following their format/style) - partial control

Final thoughts

Of course, Sphinx is a tool to build documentation. A tool does not automatically make the contents good.